

SATISH KUMAR KALLEPALLI

Staff Software Engineer • Distributed Systems • AI Platforms • Cloud-Native Backends
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PROFESSIONAL SUMMARY

Staff Software Engineer with 14+ years designing and shipping distributed backend platforms for Fortune 500 organisations (Interpublic Group, Gap Inc.). Recent work: an 80%+ manual-reduction AI rate-card pipeline, an entitlement-safe natural-language chat assistant on Snowflake Cortex, and a self-service multi-tenant investment reporting platform for IPG's media planners. Deep expertise in Java / Spring, event-driven architectures (Kafka, SQS / SNS), Snowflake data platforms, AI / LLM integrations (LangChain, LangGraph, RAG, Cortex Analyst), and Kubernetes-based observability. Technical leader who mentors distributed engineering teams, owns production incidents, and drives architecture decisions across cross-team boundaries.

SELECTED IMPACT

- **80%+ reduction** in manual rate-card processing time (AI + parallel document-extraction pipeline at IPG).
- **99.9% uptime** sustained on Halogen — IPG's flagship media operations platform — under high concurrent load.
- **40% faster deployments** after modernising CI/CD from Jenkins to GitHub Actions with parallelised pipelines.
- **1,299 tests / 0 failures** with ArchUnit-enforced package boundaries and zero-downtime blue/green EKS releases on Report Builder.
- **Millions of transactions/day** on Gap's Kafka-backed loyalty platform, Gatling-validated at peak load; centralized OIDC / OAuth 2.0 (Zuul + Apigee) standardising auth across Gap microservices.

TECHNICAL SKILLS

Languages	Java (14+ yrs), Python, JavaScript / TypeScript, SQL
Backend	Spring Boot, Spring Data JPA, Spring REST, Spring Batch, Node.js, React, Angular (signals + CDK), AngularJS, OIDC / OAuth 2.0, Zuul, Apigee, Keycloak
Distributed / Async	Apache Kafka, AWS SQS / SNS, Redis, Event-Driven Design, ExecutorService, CompletableFuture, Reactive Streams
Data	Snowflake (Semantic Views, Cortex Analyst), PostgreSQL, MongoDB, Oracle, Cosmos DB, MySQL, QuickSight, Sigma
Cloud & Infra	AWS (EC2, EKS, S3, SES, SQS, SNS, Secrets Manager), Azure, PCF, Docker, Kubernetes, Terraform
DevOps & Observability	GitHub Actions, Jenkins, Spinnaker, Maven, Gradle, Datadog, Splunk, New Relic, Micrometer, structured JSON logging
AI / LLM	Snowflake Cortex Analyst, LangChain, LangGraph, RAG, Vector DBs, Prompt Engineering, OpenAI API, MCP, AI Agents
Testing	JUnit 5, Mockito, Gatling, ArchUnit, Cucumber, Selenium
Leadership	Architecture reviews, cross-team mentoring, incident command, hiring & interviewing

PROFESSIONAL EXPERIENCE

Technical Lead / Staff Software Engineer — Kinesso (Interpublic Group), San Francisco, CA Feb 2022 – Present

AI-Powered Natural-Language Chat Assistant (Snowflake Cortex Analyst) Jul 2026 – Present

Designed and shipped an entitlement-safe NLQ interface embedded in IPG's investment reporting platform. Media planners ask business questions in English (“What's my forecast spend by market for Q3?”) and get deterministic, audit-logged Snowflake answers via Cortex Analyst.

- Eliminated ad-hoc SQL requests from planners to data engineering by architecting an entitlement-safe NL-to-SQL pipeline: HARD prompt injection of client / market entitlements, fail-closed SQL guardrail on Cortex output, one-shot retry when required filters are missing.
- Built a per-user + per-report token-bucket rate limiter (Bucket4j) with Micrometer / Datadog counters tagged by outcome and scope, plus a Flyway-migrated chat_query_audit table capturing every request, applied filters, scope, and outcome for Support and regulatory review.
- Integrated Cortex Analyst against native Snowflake SEMANTIC VIEW objects (semantic model in the Snowflake catalog rather than runtime YAML); shipped a report-scoped answer mode with a UI toggle and applied-filter chips, plus 40 golden regression questions per dataset.
- Enforced package boundaries via an ArchUnit test suite so the chat module can later be extracted to a standalone service without a rewrite; kept a green 1,299-test unit suite through every refactor.

Tech Stack: Java 21, Spring Boot, WebClient, Snowflake Cortex Analyst, Snowflake SEMANTIC VIEW, Angular signals + CDK, PostgreSQL, Flyway, Bucket4j, Micrometer, Datadog, ArchUnit, JUnit 5, Mockito, AWS Secrets Manager, JWT / RSA key-pair auth

AI-Powered Rate Card Automation Mar 2025 – Present

Architected a distributed, cloud-native AI platform transforming unstructured partner rate cards (Excel, CSV, PDF) into standardised, database-ready records — the system of record for media investment planning.

- Achieved 80%+ reduction in manual processing time by designing async, concurrent extraction pipelines with Java ExecutorService and CompletableFuture to parallelise document extraction and validation at scale.
- Engineered a high-throughput distributed data pipeline integrating Snowflake, AWS S3, and internal APIs, using Redis distributed caching to eliminate redundant lookups and cut API latency.
- Designed for horizontal scale and fault tolerance with rule-based validation and retry logic, maintaining data integrity across large batch ingestion runs; led production debugging and RCA across the pipeline, iterating on partner feedback to improve automation accuracy.

Tech Stack: Java, Spring Boot, Python, LangChain, LangGraph, RAG, Vector DBs, AI Agents, Redis, Prompt Engineering, OpenAI API, MCP, Snowflake, AWS S3, React, GitHub Actions, Datadog

Investment Reporting Platform (GATE / Report Builder) Jan 2024 – Present

Architected a self-service, multi-tenant investment reporting platform replacing legacy Excel workflows with an interactive report builder over multi-billion-dollar media commitment and forecast data in Snowflake.

- Enabled data engineering to onboard a new dataset in days rather than weeks by designing a metadata-driven query engine on Java 21 + Spring Boot, serving 5 datasets (Forecasts, Commitments, GATE 360, SPECTRA errors, Client Value) through a unified API.
- Enforced per-client / per-market row-level access on every Snowflake query via an entitlement-aware SQL generation layer with dynamic pivoting, filtering, sorting, and multi-currency conversion (IPG + GATE strategies) — defence-in-depth at both the app tier and the Snowflake role.

- Stabilised large-dataset exports that had previously OOM'd on 100K+ row reports via an async, bounded-concurrency Excel export pipeline streaming through S3 with short-transaction Postgres bookkeeping; delivered Quartz-driven scheduled exports with SES email delivery and Flyway-managed schema evolution across 45+ migrations.
- Drove blue/green EKS releases with zero-downtime deploys and owned production observability via Datadog dashboards, structured JSON logging, and MDC-tagged tracing.

Tech Stack: Java 21, Spring Boot, Spring Data JPA, Angular, AWS (EC2, EKS, S3, SES, Secrets Manager), Snowflake, PostgreSQL, Quartz, Flyway, Datadog, GitHub Actions, JUnit 5, Mockito

Halogen Media Operations Platform

Feb 2024 – Present

Led architectural modernisation of IPG's flagship Media Operations Platform — the tactical media planning lifecycle from strategy through activation and billing.

- Architected scalable distributed microservices integrating with enterprise general ledger and third-party systems, sustaining 99.9% uptime under high concurrent load; modernised CI/CD from Jenkins to GitHub Actions, cutting deployment time 40% via pipeline parallelisation.
- Directed distributed engineering teams across Mexico and Colombia through a vendor transition, and owned production reliability with Datadog dashboards and alerting — contributing to a 35% lift in user satisfaction.

Tech Stack: Java, Spring Boot, AWS EC2, RESTful APIs, MySQL, GitHub Actions, Keycloak, Apache NiFi, React, Cube.js, Datadog

Outcome Planner Strategic Application

Feb 2022 – Jan 2024

High-throughput strategic planning platform leveraging client-specific marketing mix models to optimise media spend and forecast sales outcomes at scale. Designed a distributed Spring Boot microservices architecture with an event-driven async processing layer on AWS SQS / SNS to decouple services and improve resilience under variable load. Led a distributed dev team on bi-weekly Agile releases.

Tech Stack: Java, Spring Boot, AWS (S3, SQS, SNS), RESTful APIs, MySQL, Spinnaker, JUnit, Mockito, Maven, Datadog

Lead Software Engineer — Gap Inc., San Francisco, CA

May 2018 – Jan 2022

Customer Loyalty Program

May 2020 – Jan 2022

Architected Gap's enterprise-wide loyalty platform — a high-throughput distributed system delivering real-time benefits to millions of members across all brand touchpoints.

- Designed distributed microservices (Spring Boot + REST) on Pivotal Cloud Foundry for enterprise-grade scale and fault tolerance.
- Built a high-throughput event-driven architecture on Apache Kafka for real-time point accrual and redemption across e-commerce and retail, processing millions of transactions.
- Established multi-layer performance and reliability testing (unit / service / integration + Gatling load) to validate platform behaviour under peak concurrent traffic.

Tech Stack: Java, Spring Boot, Kafka, RESTful APIs, CosmosDB, MariaDB, PCF, Swagger, JUnit, Mockito, Gradle, Gatling

OIDC / OAuth 2.0 Authentication Service

Implemented and managed a centralized OIDC / OAuth 2.0 authentication service for Gap's microservices, using Zuul Proxy for routing and Apigee for API management and security — enabling standardised authentication and authorization across the platform.

Tech Stack: OIDC, OAuth 2.0, Zuul Proxy, Apigee, Java, Spring Boot, RESTful APIs

Customer Profile Services Migration

May 2018 – Jun 2020

Led transformation of Gap's customer profile system from a legacy monolith into independent, horizontally scalable Spring Boot microservices. Executed a zero-downtime migration preserving data integrity across legacy and modern systems, and built a multi-layer test automation suite that lifted code coverage from 60% to 95%.

Tech Stack: Java, Spring Boot, RESTful APIs, CosmosDB, Oracle, PCF, JUnit, Mockito, Gradle, Gatling

Senior Software Engineer — Gap Inc. (via Nisum Technologies)

Dec 2011 – Apr 2018

Led onsite / offshore engineering teams across multiple retail platforms. Built the Price & Event Management full-stack platform on Spring Boot + MongoDB + AngularJS (PCF) — cut regression time 60% via Selenium / Protractor automation. Prior work on LOKI Localized Promotions: a Spring REST + MongoDB recommendation engine processing millions of localised pricing recommendations daily, TDD-driven with Selenium + Cucumber E2E coverage.

Tech Stack: Java, Spring 4.0 / Spring Boot, MongoDB, AngularJS, Node.js, PCF, Selenium, Cucumber, Jenkins, Gradle

EDUCATION & CERTIFICATIONS

Bachelor of Technology in Computer Science — Jawaharlal Nehru Technological University (JNTU), India 2010

Certifications: Databricks Academy Accreditation — Agentic AI Development (Apr 2026) • Full Stack Developer • Agile Awareness for Teams and Scrum Masters